

PART - A

1. Request Vote grant/deny decision :-

eg:- RequestVote (term, candidateId, lastLogIndex, lastLogTerm)

Step 1:- The server checks the term number.

3 cases:-

a) If $\text{term} < \text{currentTerm} \rightarrow \text{deny}$ (operating on old data/term)

b) $\text{term} > \text{currentTerm}$

- Update its currentTerm to
- Step down to follower state (since it was the one operating in older term)
- Treat the server as not yet having voted in this term, so votedFor is available for a new Term.

Step 2:- Check whether this server may have ^{voted} voted for that candidate ($\text{votedFor} == \text{null} \parallel \text{votedFor} == \text{candidateId}$), if not, deny the vote

Step 3:- Compare the candidate's log with the receiver's log

- at least as up-to-date $\Rightarrow$ OK
- less up-to-date $\Rightarrow$ deny the vote

Step 4:- Grant the vote, provided both checks passed

2. 'at least as up-to-date' :- $\text{candidate LastTerm} > \text{receiver LastTerm}$
or, $(\text{candidate LastTerm} = \text{receiver LastTerm} \text{ AND } \text{candidate LastIndex} > \text{receiver LastIndex})$

3. Numeric Example:-

• Receiver :

last Log Index = 6

last Log Term = 4

• Candidate :-

last Log Index = 10

last Log Term = 3

• Last Terms :-

candidate Last Term = 3

receiver Last Term = 4

∴ $3 < 4 \Rightarrow$ candidate's log is less up-to-date

Thus the receiver denies the vote

4. Split-Vote timeline :-

S1 & S2 $\Rightarrow$ synchronised election timeouts

Time (ms)	S1	S2	S3	S4	Result
0	Follower	Follower	Follower	Follower	The old leader is no longer sending heartbeats.
150	Times Out	Times Out	Waiting	Waiting	S1, S2 timeout at exactly same time.
151	Becomes candidate for term 7; votes for itself.	Becomes... (same)	Follower	Follower	Each candidate starts with 1 vote
155	Sends RequestVote to S2, S3 and S4	Sends RequestVote to S1, S3, S4	Receives competing request	Receives competing request	Requests vote across the network.
160	Receives S3's vote	Receives S4's vote	Votes for S1 first	Votes for S2 first	Each follower may vote only once in term 7.
165	Has votes from S1 & S3 : 2 votes	Has votes from S2 & S4 : 2 votes	Already voted	Already voted	Neither candidate has the required majority of 3.
300	Times out again	Times out again	Waiting	Waiting	If the retry timeouts are synchronised too, both candidates start another election together.
301	Starts term 8	Starts term 8	Follower	Follower	The same 2-2 split can happen again.

5. When randomized timeouts breaks the symmetry

S1 $\Rightarrow$ 173 ms

S2 $\Rightarrow$ 241 ms

Time after term 7 split
(ms)

S1

S2

What happens

0

Chooses ~~173~~
173 ms timeout

Chooses 241 ms
timeout

Randomized $\Rightarrow$
candidates' deadlines
different

173

Times out 1st &
starts term 8

Still waiting

Symmetry breaks
here. Only S1 is
campaigning.

180

Requests votes from
S2, S3 and S4

Receives S1's
highest log-term
request while not yet
campaigning

S1 can collect
votes ~~but~~
without racing against
S2.

190

Receives a majority
and becomes leader

Remains or becomes
followers

S1 wins term 8.

195

Sends AppendEntries
heartbeats

Receives heartbeat
and resets its
election timer

S2 no longer
reaches its original
241 ms election
deadline